A proof of the conjectured closed form for OEIS A366932

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Abstract

OEIS A366932 carries a conjectured closed form for its terms. We prove it. The entry also posts a generating function $A(x)$; the conjecture is then exactly the statement that the generating function of the conjectured formula agrees with A. The formula is a $\mathbb{Q}$-linear combination of terms $n^k r^n$, whose generating function is computed in closed form by applying $p(\theta)$ to $1/(1-rx)$, with $\theta = x\,\frac{d}{dx}$. The difference of the two generating functions is then an explicit rational function; here it collapses to the polynomial 0, of degree 0, which proves the closed form for every $n > 0$. The computation is exact rational-function arithmetic, so the conclusion holds for all n at once rather than for the terms inspected.

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1 The sequence and the conjecture

OEIS A366932 is "Total number of edges formed after n points have been placed in general position on each edge of a triangle (as in A365929)". It has offset 0 and begins

$$a(0), \dots, a(7) \;=\; 3, 9, 51, 255, 855, 2193, 4719, 8991,\; \dots$$

The entry gives the generating function

$$A(x) \;=\; \sum_{n\geq 0} a(n)\, x^n \;=\; -\frac{3\,(x+1)\left(5x^3+15x^2-3x+1\right)}{(x-1)^5}, \tag{1}$$

and it carries the following comment:

> **Conjecture:** a(n) = 3*(3*n^4/2 - 2*n^3 + 3*n^2/2 + n + 1). [This is now a theorem - *N. J. A. Sloane*, Dec 31 2025]

As of the "Last modified" line on the live entry (2026-04-17, revision 111) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Write $f(n)$ for the conjectured closed form,

$$f(n) \;=\; \frac{9n^4}{2} - 6n^3 + \frac{9n^2}{2} + 3n + 3.$$

2 Closed forms as generating functions

Let $A(x) = \sum_{n\geq 0} a(n)x^n$ and let $\theta = x\,\dfrac{d}{dx}$, so θ acts on x^m as multiplication by m.

Lemma 1. *Let $r \in \mathbb{Q}$ and let p be a polynomial. Then*

$$\sum_{n\geq 0} p(n)\, r^n x^n \;=\; \left(p(\theta)\tfrac{1}{1-rx}\right)(x),$$

a rational function of x. Consequently the generating function of any $\mathbb{Q}$-linear combination of terms $n^k r^n$ is rational and computable in closed form.

Proof. $\sum_{n\geq 0} r^n x^n = 1/(1-rx)$, and applying θ multiplies the coefficient of x^n by n; so θ^k multiplies it by n^k, and linearity gives the claim. Starting the sum at an offset c instead of 0 subtracts the polynomial $\sum_{n<c} p(n) r^n x^n$, which changes nothing below. □

Corollary 2. *Let $F(x) = \sum_{n\geq 0} f(n)x^n$. Then $f(n) = a(n)$ for every $n > d$ if and only if $F - A$ is a polynomial of degree at most d.*

Proof. The coefficient of x^n in $F - A$ is $f(n) - a(n)$, so that difference vanishes for all $n > d$ exactly when $F - A$ has no terms above x^d. □

3 The computation for A366932

By Lemma 1 the conjectured formula has generating function

$$F(x) \;=\; \frac{3\left(-5x^4 - 20x^3 - 12x^2 + 2x - 1\right)}{x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1},$$

and subtracting the entry's generating function (1) gives, after exact cancellation of rational functions,

$$F(x) - A(x) \;=\; 0,$$

a polynomial of degree 0.

Theorem 3. *For every $n > 0$,*

$$a(n) \;=\; \frac{9n^4}{2} - 6n^3 + \frac{9n^2}{2} + 3n + 3.$$

Proof. Immediate from Corollary 2 and the displayed value of B. □

Remark 4. The polynomial $F - A$ is not an error term: it records the boundary of the claim. A closed form of this kind typically fails at a few initial indices, where the sequence's own definition has not yet settled into its eventual pattern, and $F - A$ collects exactly those discrepancies. Above that range the identity is exact.

4 Verification

Every number above is an output of a script written separately from this note, and two independent checks were run.

First, the generating function was checked against the entry rather than assumed: the Taylor coefficients of (1) were expanded and compared term by term with the DATA section of A366932, agreeing on all 13 terms compared.

Second, the closed form was evaluated directly on the published terms, without reference to any generating function: $f(n)$ was computed in exact rational arithmetic and compared with the entry's own value for every $n > 0$ in range, agreeing in all 43 cases from $n = 0$ upward. This is what Corollary 2 predicts from the degree of $F - A$, and it is a genuinely different computation from the symbolic one.

The symbolic step is exact throughout. The difference is obtained by rational-function cancellation in K, not by series truncation, so the identity $B = 0$ is an identity of functions and the theorem holds for all $n > 0$ simultaneously.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A366932.

[2] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011, Chapter 7 (holonomic closure properties and the translation between recurrences and differential equations).

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999, Chapter 6 (algebraic and D-finite generating functions).